Fiscal Year:	FY 2004	**Task Last Updated:**	FY 07/24/2006
PI Name:	Heesch, Cheryl Miller Ph.D.		
Project Title:	Gender Related Differences In Central Autonomic Control Following Simulated Microgravity In Rats		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Organismal biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Developmental Biology (2) Musculoskeletal Biology (3) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	65211	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Space Biology 01-OBPR-06
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA ARC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Hasser, Eileen Ph.D. (University of Missouri/Dalton Cardiovascular Res. Ctr.) Foley, Charles Ph.D. (University of Missouri/Dalton Cardiovascular Res. Ctr.)		
Grant/Contract No.:	NAG2-1603		
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Task Description:	Prolonged bedrest or exposure to reduced gravity, as occurs during space flight, results in cardiovascular deconditioning. In astronauts, upon returning to the normal gravity of earth, there is a decreased ability to exercise and a greater tendency to faint when standing. Normally, when a person stands up, heart rate increases and increased nerve activity (sympathetic nerve activity) to blood vessels causes vasoconstriction, so that blood is returned to the heart and doesn't pool in the lower extremities. Following cardiovascular deconditioning the normal compensatory vasoconstriction is not as pronounced, and blood isn't returned to the heart as effectively. This can result in light-headedness and fainting. Interestingly, following space flight this tendency to faint upon standing is more pronounced in female, compared to male, astronauts. Previously, using an animal model for cardiovascular deconditioning, the hindlimb unloaded rat, we demonstrated that in both males and females the ability to increase sympathetic nerve activity when blood pressure drops is attenuated following deconditioning (14 days of hindlimb unloading). When age-matched male and female control animals were evaluated, we found that, compared to males, female rats exhibited less of a compensatory increase in vasoconstrictor sympathetic nerve activity when blood pressure was lowered. Thus, the combined effect of sex and cardiovascular deconditioning resulted in less ability to defend against a drop in blood pressure in the females. Since baseline heart rate is higher in females compared to males, and since cardiovascular deconditioning results in an increase in heart rate, we considered that heart rate might be different between males and females and might be affected differently by cardiovascular deconditioning. Heart rate is determined by the balance between sympathetic nerve activity to the heart, which would increase heart rate, and parasympathetic nerve activity, which slows the heart. In control rats, females had a higher resting heart rate compared to males and the resting heart rate in females was more dependent on sympathetic nerve activity to the heart. Baseline heart rate increased in both male and female rats following cardiovascular deconditioning and it was due to both decreased parasympathetic and increased sympathetic influences on the heart in both sexes. However, cardiovascular deconditioning did not affect the compensatory increase in heart rate when blood pressure was decreased in either males or females. Therefore, it appears that in response to cardiovascular deconditioning, a greater attenuation of compensatory vasoconstriction in females compared to males, is associated with attenuated increases in sympathetic nerve activity to the blood vessels. When blood pressure drops, during an orthostatic challenge or due to hemorrhage, humoral systems also contribute to the compensatory response. Vasopressin, a hormone which is released when blood pressure drops, has direct effects on blood vessels to cause constriction and therefore might work to counteract the initial decrease in blood pressure. We found that baseline levels of vasopressin are elevated in hindlimb unloaded rats, and this effect is greater in males compared to females. In addition, when rats are exposed to progressive hemorrhage, the progressive decrease in blood pressure is accompanied by a progressive elevation in plasma vasopressin Orthostatic intolerance, or the inability to maintain adequate cardiac output and cerebral blood flow while standing, is a common feature of cardiovascular deconditioning following spaceflight or prolonged bedrest. Interestingly, the incidence of presyncope (lightheadedness, loss of peripheral vision, sudden drop in systolic pressure) following space flight is much greater in female compared to male astronauts. Alterations in blood volume, cardiac function, vascular function, and neurohumoral control mechanisms have all been implicated in orthostatic intolerance following cardiovascular deconditioning. Normal adjustments to changes in posture require appropriate neurohumoral control mechanisms, which likely differ in females compared to males. Hindlimb unloading (HU) in rodents is an animal model of cardiovascular deconditioning, and produces cardiovascular effects similar to those observed in humans. No studies have critically evaluated potential differences in the central nervous mechanisms controlling autonomic function that may contribute to sex differences in cardiovascular deconditioning. The general hypothesis to be tested is that: Compared to male rats, female rats will exhibit greater signs of orthostatic intolerance following hindlimb unloading. These effects are centrally mediated, involve the neurotransmitter GABA in central autonomic pathways, and differences are associated with ovarian hormone levels.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	During the first year of this project we reported that arterial baroreflex control of renal sympathetic nerve activity (RSNA) was attenuated in females compared to males. Hindlimb unloading (HU) also blunted baroreflex control of RSNA in both sexes. These changes in the baroreflex curve were reflected by reduced maximum sympathoexcitation and reduced slope (sensitivity) of arterial baroreflex control of efferent sympathetic nerve activity. We conclude that cardiovascular deconditioning attenuates baroreflex mediated increases in sympathetic activity in both males and females. The additive effect of deconditioning and sex reduces sympathetic reserve to a greater extent in females as compared to males. This blunted sympathetic reserve could contribute to the greater incidence of orthostatic intolerance in females following cardiovascular deconditioning.
Bibliography Type:	Description: (Last Updated: 01/15/2021)
Abstracts for Journals and Proceedings	Heesch, C.M.; Foley, C.M.; Phaup, G.J.; Hasser, E.M. "Gender differences in autonomic determinants of heart rate (HR) in cardiovascular (CV) deconditioning." Experimental Biology '04 Meeting, March 2004. FASEB Journal 2004 Mar;18: A1076. , Mar-2004
Abstracts for Journals and Proceedings	Foley, C.M.; Price, E.M.; Heesch, C.M. "GAGAA receptor alpha 1 and alpha 2 mRNA expression in nucleus tractus solitarius in nonpregnant and pregnant rats." Experimental Biology '04 Meeting, March 2004. FASEB Journal 2004 Mar;18: A1078. , Mar-2004
Abstracts for Journals and Proceedings	Haught, A.L.; Foley, C.M.; Mueller, P.J., Friskey, S.A.; Rossi, N.F.; Heesch, C.M.; Hasser, E.M. "Effects of cardiovascular deconditioning on the neurohumoral response to blood loss." Experimental Biology '04 Meeting, March 2004. FASEB Journal 2004 Mar;18: A1073. , Mar-2004

Abstracts for Journals and Proceedings	Martins-Pinge, M.C.; Mueller, P.J.; Foley, C.M.; Friskey, S.A.; Heesch, C.M.; Hasser, E.M. "Cardiovascular effects of nitric oxide and GABA in the paraventricular nucleus of the hypothalamus in conscious rats." Experimental Biology '04 Meeting, March 2004. FASEB Journal 2004 Mar;18: A668. , Mar-2004